Project 2 : WSOP Main Event data analysis

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Workflow & Reproducibility Guide

To ensure this entire workflow is fully reproducible, follow these steps:

Download the raw dataset from Kaggle (link provided in the Sources section).

Place the file WSOP___1971-2025.csv in your working directory.

Run this Quarto document. The R setup will automatically clean the data, normalize it into 3rd Normal Form (3NF) relational tables, export them as temporary CSVs, and ingest them into a local DuckDB file (poker_analysis.duckdb).

Accessing data & Designing the database

```
wsop_old <- read_csv("WSOP__1971-2025.csv")
```

```
Rows: 20083 Columns: 12
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (4): name, city, state, country
dbl  (6): year, place, prize, entries, prizepool, buyin
date (2): start, finish

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
wsop_raw <- wsop_old %>%
  mutate(
    country = case_when(
      country %in% c("US", "USA") ~ "USA",
      is.na(country) ~ "Unknown",
      TRUE ~ country
    ),

    city = replace_na(city, "Unknown"),
    state = replace_na(state, "Unknown")
  )

# Players

players <- wsop_raw %>%
  select(name, city, state, country) %>%
  distinct() %>%
  mutate(player_id = row_number()) %>%
  relocate(player_id)

# Tournaments
tournaments <- wsop_raw %>%
  select(year, entries, prizepool, buyin, start, finish) %>%
  distinct() %>%
  mutate(event_id = row_number()) %>%
  relocate(event_id)

# Results
results <- wsop_raw %>%
  left_join(players, by = c("name", "city", "state", "country")) %>%
  left_join(tournaments, by = c("year", "entries", "prizepool", "buyin", "start", "finish"))
  select(player_id, event_id, place, prize) %>%
  mutate(result_id = row_number()) %>%
  relocate(result_id)


write_csv(players, "players.csv")
write_csv(tournaments, "tournaments.csv")
write_csv(results, "results.csv")

# DuckDB
con <- dbConnect(duckdb(), dbdir = "poker_analysis.duckdb")
```

```r
# --- Players ---
dbExecute(con, "DROP TABLE IF EXISTS players;")
```

```
[1] 0
```

```r
dbExecute(con, "CREATE TABLE players (
    player_id INTEGER PRIMARY KEY,
    name VARCHAR(255),
    city VARCHAR(255),
    state VARCHAR(50),
    country VARCHAR(100)
);")
```

```
[1] 0
```

```r
dbExecute(con, "COPY players FROM 'players.csv' (FORMAT CSV, HEADER);")
```

```
[1] 14544
```

```r
# --- Tournaments ---
dbExecute(con, "DROP TABLE IF EXISTS tournaments;")
```

```
[1] 0
```

```r
dbExecute(con, "CREATE TABLE tournaments (
    event_id INTEGER PRIMARY KEY,
    year INTEGER,
    entries INTEGER,
    prizepool BIGINT,
    buyin INTEGER,
    start DATE,
    finish DATE
);")
```

```
[1] 0
```

```r
dbExecute(con, "COPY tournaments FROM 'tournaments.csv' (FORMAT CSV, HEADER);")
```

```
[1] 54
```

```r
# --- Results ---
dbExecute(con, "DROP TABLE IF EXISTS results;")
```

```
[1] 0
```

```r
dbExecute(con, "CREATE TABLE results (
    result_id INTEGER PRIMARY KEY,
    player_id INTEGER,
    event_id INTEGER,
    place INTEGER,
    prize BIGINT
);")
```

```
[1] 0
```

```r
dbExecute(con, "COPY results FROM 'results.csv' (FORMAT CSV, HEADER);")
```

```
[1] 20083
```

```r
dbGetQuery(con, "SELECT * FROM players LIMIT 5;")
```

```
  player_id                   name        city state country
1         1            Johnny Moss      Dallas    TX     USA
2         2 Amarillo Slim Preston    Amarillo    TX     USA
3         3          Puggy Pearson   Las Vegas    NV     USA
4         4    Crandall Addington San Antonio    TX     USA
5         5          Bryan Roberts     Houston    TX     USA
```

Querying data using SQL

Global Dominance: Top 10 Countries by Total WSOP Winnings

Analyzing which countries have produced the highest cumulative prize money through successful tournament finishes.

```
SELECT
    p.country,
    SUM(r.prize) AS total_winnings,
    COUNT(r.result_id) AS total_cashes
FROM players AS p
JOIN results AS r ON p.player_id = r.player_id
WHERE p.country IS NOT NULL
GROUP BY p.country
ORDER BY total_winnings DESC
LIMIT 10;
```

Table 1: Displaying records 1 - 10

country	total_winnings	total_cashes
USA	901341283	12216
CA	74109810	910
GB	58676706	783
Unknown	56814263	1481
DE	38766389	355
FR	34153251	424
SE	25738211	138
AU	23788729	164
NO	18686657	77
IT	17390059	146

#Top 10 Career Earners in Main Event History #Identifying the individual players who have secured the most prize money across their entire WSOP careers.

```sql
SELECT
    p.name,
    SUM(r.prize) AS career_earnings
FROM players AS p
JOIN results AS r ON p.player_id = r.player_id
GROUP BY p.name
ORDER BY career_earnings DESC
LIMIT 10;
```

Table 2: Displaying records 1 - 10

name	career_earnings
Michael Mizrachi	12423169
Daniel Weinman	12159200
Jamie Gold	12017500
Joe Cada	10728068
Jonathan Tamayo	10525789
Espen Jorstad	10015000
Martin Jacobson	10015000
Hossein Ensan	10000000
John Cynn	9475500
Peter Eastgate	9221395

#Top 10 Highest ROI Performances # Calculating the Return on Investment (ROI) to determine which tournament runs were the most profitable relative to the buy-in.

```sql
SELECT
    t.year,
    t.buyin,
    r.prize,
    ((r.prize - t.buyin) * 100.0 / t.buyin) AS roi_percentage
FROM tournaments AS t
JOIN results AS r ON t.event_id = r.event_id
WHERE r.prize > 0
ORDER BY roi_percentage DESC
LIMIT 10;
```

Table 3: Displaying records 1 - 10

year	buyin	prize	roi_percentage
2023	10000	12100000	120900.00
2006	10000	12000000	119900.00
2022	10000	10000000	99900.00
2014	10000	10000000	99900.00
2025	10000	10000000	99900.00
2024	10000	10000000	99900.00
2019	10000	10000000	99900.00
2008	10000	9152416	91424.16
2010	10000	8944310	89343.10
2018	10000	8800000	87900.00

Most Reliable Players with 10+ Career Cashes

Filtering for elite players who consistently finish in the money, demonstrating long-term skill and stability.

```sql
SELECT
    p.name,
    COUNT(r.result_id) AS cash_count,
    AVG(r.place) AS average_place
FROM players AS p
JOIN results AS r ON p.player_id = r.player_id
GROUP BY p.name
HAVING COUNT(r.result_id) >= 10
ORDER BY average_place ASC
LIMIT 10;
```

Table 4: 3 records

name	cash_count	average_place
Berry Johnston	10	24.0000
Johnny Chan	11	297.0909
Allen Cunningham	11	443.3636

Year-over-year growth in entries

```sql
SELECT
    year,
    SUM(entries) AS total_entries,
    LAG(SUM(entries)) OVER (ORDER BY year) AS previous_year_entries,
    ROUND(
        100.0 * (SUM(entries) - LAG(SUM(entries)) OVER (ORDER BY year))
        / LAG(SUM(entries)) OVER (ORDER BY year), 2
    ) AS yoy_entry_growth_pct
FROM tournaments
GROUP BY year
ORDER BY year;
```

Table 5: Displaying records 1 - 10

year	total_entries	previous_year_entries	yoy_entry_growth_pct
1971	6	NA	NA
1972	8	6	33.33
1973	13	8	62.50
1974	16	13	23.08
1975	21	16	31.25
1976	22	21	4.76
1977	34	22	54.55
1978	42	34	23.53
1979	54	42	28.57
1980	73	54	35.19

Top 10 years by total prizepool

```sql
SELECT
    year,
    SUM(prizepool) AS total_prizepool,
    SUM(entries) AS total_entries,
    COUNT(*) AS total_events
FROM tournaments
GROUP BY year
```

```
ORDER BY total_prizepool DESC
LIMIT 10;
```

Table 6: Displaying records 1 - 10

year	total_prizepool	total_entries	total_events
2024	94041600	10112	1
2023	93399900	10043	1
2025	90535500	9735	1
2006	82512162	8773	1
2022	80782475	8663	1
2019	80548604	8569	1
2018	74015600	7874	1
2010	68799059	7319	1
2017	67877400	7221	1
2011	64540858	6865	1

Buy-in distribution by year

```
SELECT
    CASE
        WHEN buyin < 1000 THEN 'Under $1K'
        WHEN buyin BETWEEN 1000 AND 4999 THEN '$1K-$4,999'
        WHEN buyin BETWEEN 5000 AND 9999 THEN '$5K-$9,999'
        ELSE '$10K+'
    END AS buyin_tier,
    COUNT(*) AS number_of_tournaments,
    AVG(entries) AS avg_entries,
    AVG(prizepool) AS avg_prizepool
FROM tournaments
GROUP BY buyin_tier
ORDER BY number_of_tournaments DESC;
```

Table 7: 2 records

buyin_tier	number_of_tournaments	avg_entries	avg_prizepool
$10K+	53	3005.887	28247779

buyin_tier	number_of_tournaments	avg_entries	avg_prizepool
$5K-$9,999	1	6.000	30000

Most Profitable Buy-In Levels

```
SELECT
    t.buyin,
    COUNT(r.result_id) AS total_cash_results,
    ROUND(AVG(r.prize), 2) AS avg_prize,
    MAX(r.prize) AS max_prize,
    SUM(r.prize) AS total_prize_awarded
FROM tournaments AS t
JOIN results AS r
    ON t.event_id = r.event_id
WHERE t.buyin IS NOT NULL
GROUP BY t.buyin
HAVING COUNT(r.result_id) >= 20
ORDER BY avg_prize DESC
LIMIT 10;
```

Table 8: 1 records

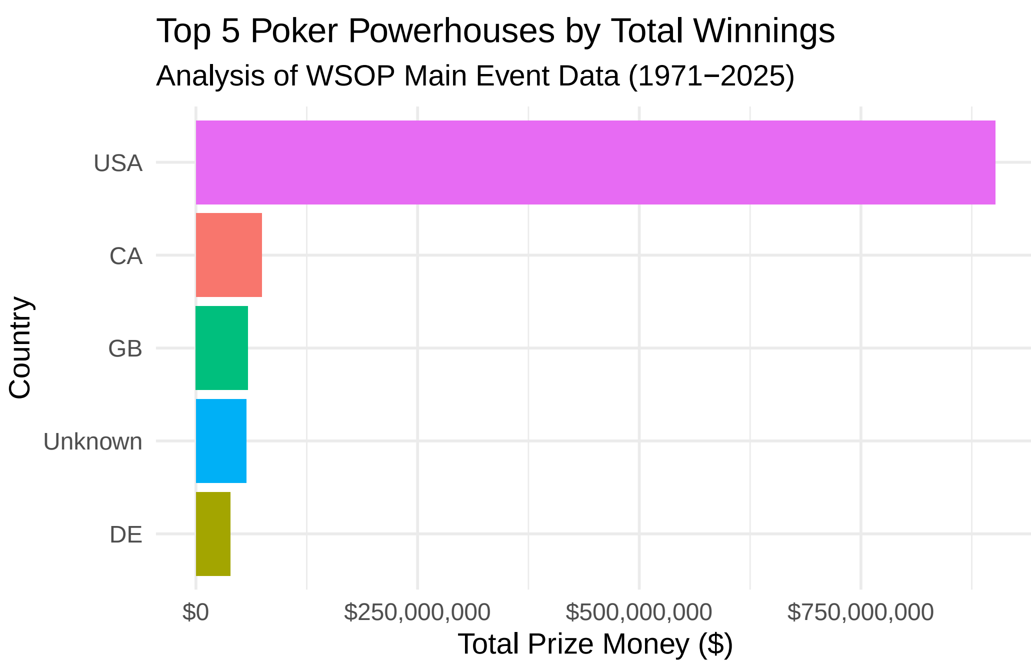
buyin	total_cash_results	avg_prize	max_prize	total_prize_awarded
10000	20082	74463	12100000	1495365929

Creating data visualizations

```
country_data <- dbGetQuery(con, "
    SELECT p.country, SUM(r.prize) as total
    FROM players AS p
    JOIN results AS r ON p.player_id = r.player_id
    WHERE p.country IS NOT NULL
    GROUP BY p.country
```

```
    ORDER BY total DESC
    LIMIT 5")

ggplot(country_data, aes(x = reorder(country, total), y = total, fill = country)) +
  geom_col(show.legend = FALSE) +
  scale_y_continuous(labels = scales::dollar_format()) +
  coord_flip() +
  labs(
    title = "Top 5 Poker Powerhouses by Total Winnings",
    subtitle = "Analysis of WSOP Main Event Data (1971-2025)",
    x = "Country",
    y = "Total Prize Money ($)"
  ) +
  theme_minimal()
```

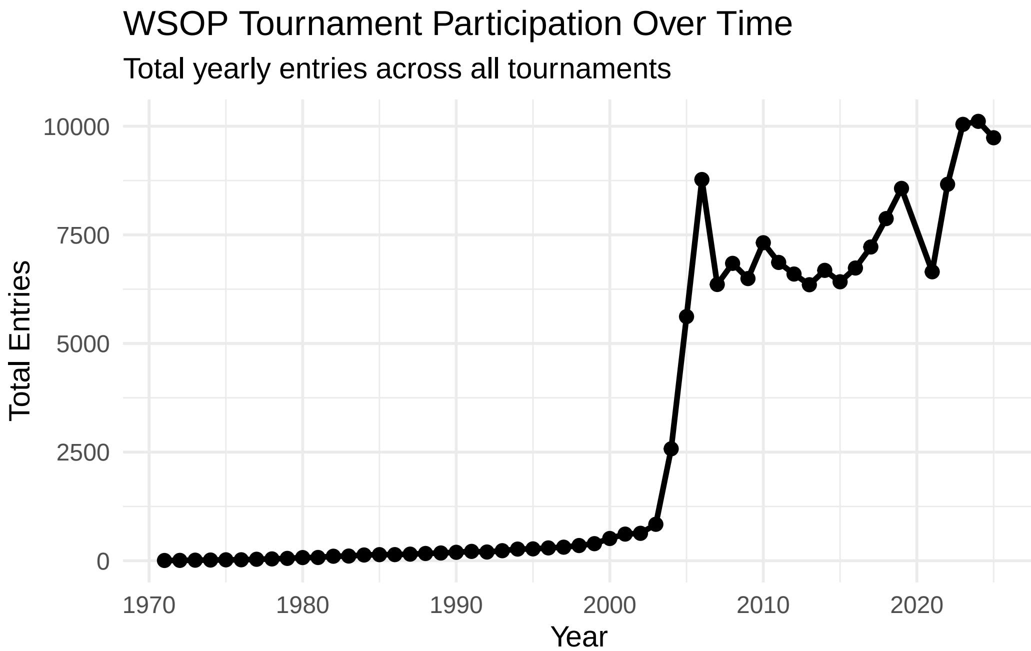


YoY winning table

```
yoy_growth <- dbGetQuery(con, "
SELECT
    year,
```

```
    SUM(entries) AS total_entries
FROM tournaments
GROUP BY year
ORDER BY year
")

ggplot(yoy_growth, aes(x = year, y = total_entries)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(
    title = "WSOP Tournament Participation Over Time",
    subtitle = "Total yearly entries across all tournaments",
    x = "Year",
    y = "Total Entries"
  ) +
  theme_minimal()
```




```
# terminate the SQL connection

DBI::dbDisconnect(con, shutdown = TRUE)
```

Sources

- https://www.kaggle.com/datasets/cviaxmiwnptr/wsop-main-event-results-1971-2024?resource=download
- web link/document title/LLM prompt